

Space-time Estimation Techniques For UTRA System

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ABSTRACT

This paper investigates the space-time channel estimation techniques in a multiple-access communication receiver system that receives coded digital data modulated using DS-CDMA, with an antenna array in a multipaths propagation environment. A spatial and temporal channel model has been implemented, which is built upon the understanding of multipath fading and Doppler spread with merging time delay spread, direction-of-arrival and adaptive antenna array geometry. A receiver structure based on space-time processing has also been implemented, i.e. a reverse link, single-user (SU) with single antenna input (SI) at mobile and multiple antenna output (MO) at the base (SU-SIMO). A reverse link, multiuser (MU) – SIMO has been presented as well. This paper also provides a brief overview of various methods available for estimation of DOA of a radio signal using an antenna array. Joint Angle and Delay Estimation (JADE) combined with MUSIC is proposed as the main candidate for estimating the channel profile. For this purpose we present the performance of MUSIC as used in UTRA system.

INTRODUCTION

The next generation mobile communication systems, called Universal Mobile Telecommunication System (UMTS) or International Mobile Telecommunication – 2000 (IMT-2000) system in the International Telecommunication Union (ITU), must be flexible enough to support wideband services with the same quality as in fixed network.

Direct Sequence CDMA technology is considered to be the promising candidate for UMTS wireless access because of its numerous advantages, in addition to its interference immunity and bandwidth efficiency. This paper investigates a spatial and temporal communication system based on an adaptive antenna array, which includes spatial and temporal channel modeling, optimum detection for single-user and multi-users CDMA. Such spatial and temporal processing provides significant improvement of performance against multipath fading in mobile radio communications. Paulraj (1), Kohno (2).

The Direction-of-Arrival (DOA) estimation techniques play an important role in the mobile communication system. In addition to the well-known and expected capacity enhancement that can be obtained by exploiting the DOA of the signal, It is also the method

used to determine the position of a mobile user, which will be important for emergency services.

SYSTEM MODEL

In general we define *Adaptive Antenna* as an array of antennas which is able to change its antenna patterns dynamically to adjust to noise, interference, and multipath. Adaptive array antennas can adjust their pattern to track mobile users. They are used to enhance received signals and may also be used to form beams for transmission. *Switched beam systems* use a number of fixed beams at an antenna site. The receiver selects the beam that provides the greatest signal enhancement and interference reduction. Switched beam systems may not offer the degree of performance improvement offered by adaptive systems, but they are often much less complex and are easier to retro-fit to existing wireless Technologies. *Smart antenna systems* can include both adaptive antenna and switched beam technologies. Smart antennas use an array of low gain antenna elements which are connected by a combining network. An arbitrary array of elements is shown in Fig 1. Here, θ is the azimuthal angle and ϕ is the elevation angle of a plane wave incident on the array. The horizon is represented by $\phi = \pi/2$.

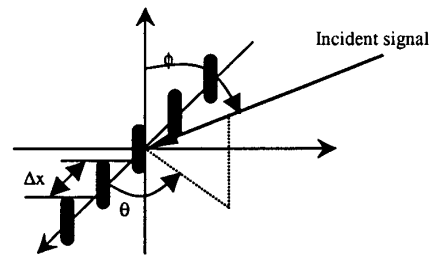


Figure 1: Simple array of antenna elements.

To simplify the analysis of antenna arrays, we make the following assumptions. The spacing between array elements is small enough that there is no amplitude variation between the signals received at different elements. There is no mutual coupling between elements. All incident fields can be decomposed into a discrete number of plane waves. That is, there are a finite number of signals. The bandwidth of the signal incident on the array is small compared with the carrier frequency.

The array may consist of a number of antenna elements distributed in any desired pattern; however, the array is frequently implemented as a linear equally spaced

(LES), uniform circular or uniformly spaced planar array of similar, co-polarised, low gain elements which are oriented in the same direction. A simple M -element LES antenna array, oriented along the x -axis, with an element spacing of Δx , is illustrated in Fig 2.

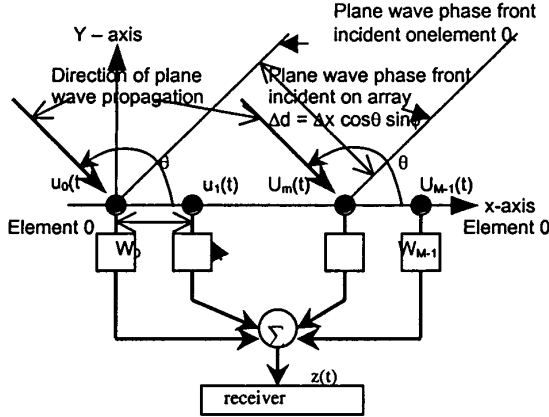


Figure 2: A baseband complex envelope model of a LES array oriented along the x -axis, receiving a plane wave from direction (ϕ, θ) .

The baseband complex envelope representation of the array is shown in Fig 2. Each branch of the array has a weighting element, W_m . The Weighting element, W_m , has both a magnitude and a phase associated with it. Consider a plane wave incident on the array from an angle (ϕ, θ) relative to the axis of the array. We represent the modulation of the plane wave using the baseband complex envelope, $S(t)$. Assume for the moment that all of the array elements are noiseless isotropic antennas which have uniform gain in all directions. With $X_m = m\Delta x$, the signal received at an antenna element m on the LES is

$$\begin{aligned} U_m(t) &= A s(t) e^{-j\beta_m \Delta x} \\ &= A s(t) e^{-j\beta_m \Delta x \cos \theta \sin \phi} \end{aligned}$$

where $\beta = 2\pi / \lambda$ is the phase propagation factor, $\lambda = f/c$, λ is the wavelength, c is speed of light, f is carrier frequency and A is an arbitrary gain constant.

The signal $Z(t)$ at the array output is

$$\begin{aligned} Z(t) &= \sum_{m=0}^{M-1} W_m U_m(t) \\ &= A s(t) \sum_{m=0}^{M-1} W_m e^{-j\beta_m \Delta x \cos \theta \sin \phi} \\ &= A s(t) f(\phi, \theta) \end{aligned}$$

The term $f(\phi, \theta)$ is called the array factor. The array factor determines the ratio of the received signal available at the array output, $Z(t)$, to the signal, $A s(t)$, measured at the reference element as a function of Direction-of-Arrival; (ϕ, θ) . By adjusting the set of weights, $\{W_m\}$, it is possible to direct the maximum of the main beam of the array factor in any desired direction, (ϕ_0, θ_0) .

The received power at the array output is:

$$P_r = \frac{1}{2} |Z(t)|^2 = \frac{1}{2} |A s(t)|^2 |f(\phi, \theta)|^2$$

To show how the weights, $\{W_m\}$, can be used to change the antenna pattern of the array, let the M^{th} Weight be given by

$$W_m = e^{j\beta_m \Delta x \cos \theta}$$

In the general case, the array factor pattern is a function of both θ and ϕ . If the field pattern of each array element is $g_a(\phi, \theta)$ and all elements are identical and oriented in the same direction, then the total field pattern of the array is given by

$$F(\phi, \theta) = f(\phi, \theta) g_a(\phi, \theta)$$

This is called the principle of pattern multiplication Stutzman (3).

Each element also has a polarisation state associated with it, as does any incident plane wave. It is assumed that the plane waves incident on each element are co-polarised so that there is no polarisation loss in the interaction between the plane wave and each element.

In working with array antennas, it is very convenient to make use of vector notation. We define the weight vector as

$$W = [W_0 \ \dots \ W_{M-1}]^H$$

Where the superscript H represents the Hermitian transpose.

The signals from each antenna elements are grouped in a data vector

$$U = [U_0 \ \dots \ U_{M-1}]^T$$

Then the array output, $Z(t)$, can be expressed as the inner product of the array weight vector, W , and the data vector, $U(t)$,

$$Z(t) = W^H U(t)$$

The array factor in a direction (ϕ, θ) is

$$f(\phi, \theta) = W^H a(\phi, \theta)$$

The vector $a(\phi, \theta)$ is called the steering vector in direction (ϕ, θ) . Given a plane wave incident from a direction (ϕ, θ) as illustrated in Fig 2, the steering vector, $a(\phi, \theta)$ describes the phase of the signal available at each antenna element relative to the phase of the signal at the reference element (element 0). The steering vector is

$$a(\phi, \theta) = [1 \ a_1(\phi, \theta) \ \dots \ a_{M-1}(\phi, \theta)]^T$$

A set of steering vectors, either measured or calculated over all values of θ and ϕ , is called the array manifold. In addition to being useful in the analysis of arrays, knowledge of the array manifold is critical in direction finding, downlink beamforming, and other aspects of array operation. The angle pair (ϕ, θ) is called the Direction-of-arrival (DOA) of the received plane wave.

Model for Space-Time Processing

Basically there are two models for the space-time processing applications in the reverse link Paulraj (1), i.e. where we do not consider the Multiple Inputs Multiple Output at the base (MIMO). (i) Single User (SU) with Single Antenna Input (SI) at mobile and Multiple Antenna Output (MO) at the base station. (ii) Multiuser (MU) with Single Antenna Input (SI) at each mobile and Multiple Antenna Composite Output (MO) at the base.

The block diagram of signal model that applied in the simulation is shown in Fig 3. This model is based on the knowledge of the characteristics of Space-time Radio channel.

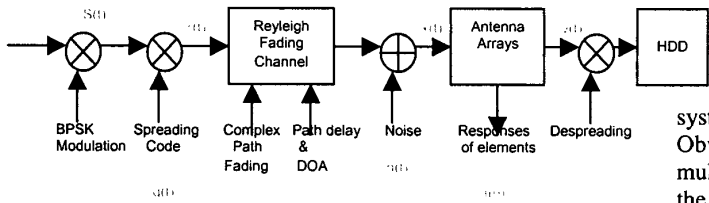


Figure 3: Block diagram of the simulated system

In this model, two users are assumed communicating simultaneously. The multipath signal of user 1 arrives at 0° (direct path), 30° and 60° , with zero delay, one chip (T_c) and two chip period ($2T_c$) delayed respectively. It assumes an angle spread of 60° . Similar assumption is made to the User 2 as well, with arrival angles of 160° (direct path), 180° (one T_c delay) and 200° (two T_c delays). This is illustrated in the Fig 4.

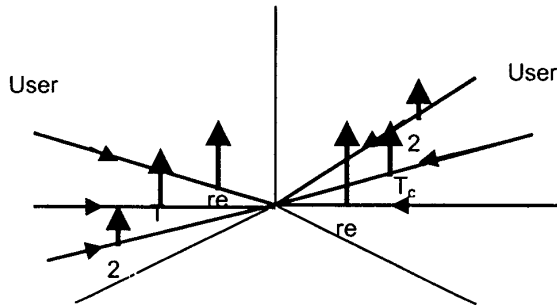


Figure 4: 3D Profile (DOA, Delay & Amplitude) of a two users multipath channel.

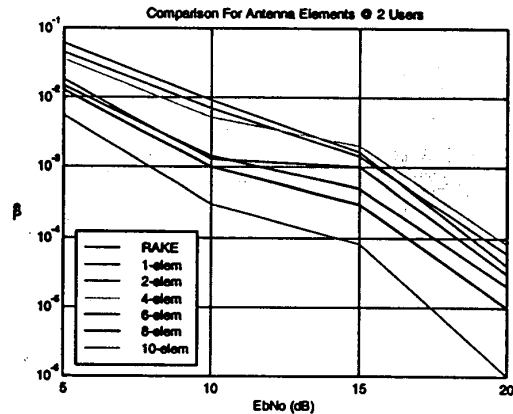


Figure 5: Antenna Array Performance vs. no. of elements.

Fig 5 shows the performance of the antenna array for different number of antenna elements. It is clear that the BER performance is significantly improved as the number of elements increases. The performance of 10

elements gives about 8 dB gain over the 1 element antenna.

In simulation it is assumed that there are two users, and that 16 orthogonal Walsh code are used. A similar assumption is also made for the RAKE receiver.

MAI cancellation was not performed; its performance is more competent than a RAKE receiver that has MAI cancellation when the system is applied with about 6 antenna elements. Obviously, the above result shows that if we employ multipath diversity for MAI cancellation in this model, the performance of this model will be much better than a RAKE receiver as is shown in Nocolacacos (4). This is because the antenna array can reduce the interference by using the antenna directivity.

Fig 6 illustrates the effect of DOAs in a multipath channel arriving at a base station. The red line (different DOAs) is the simulation result of the model shown in Fig 4. i.e. both users have their own DOAs without interfering with each other. On the other hand the blue line (same DOAs) assumes that both users have the same DOAs (0° , 30° and 60°).

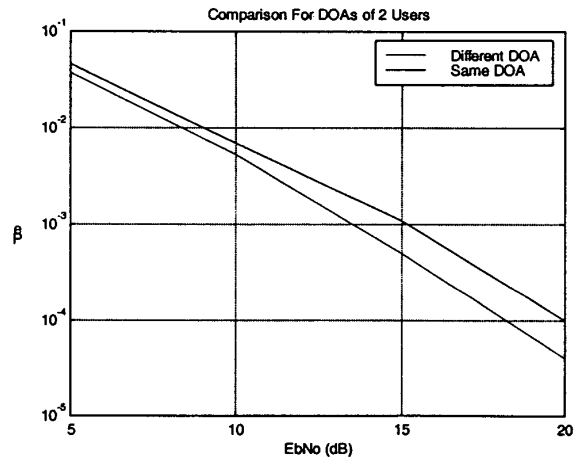


Figure 6: Performance for different DOA for two users signal.

If the DOAs for both users arrive in the same directions, this will cause more interference, consequently the receiver would not be able to distinguish the users easily. This is why Fig 6 shows that the BER for the same DOA is about 2 dB higher than the case with different DOA.

CHANNEL ESTIMATION ALGORITHMS

Recent research on the spatial and temporal signal processing using an adaptive antenna array has been widely studied. Generally, the channel estimation techniques can be classified into three approaches. Least Mean Squares (LMS), Recursive Least Squares (RLS) and Constant Modulus Algorithm (CMA) are the adaptive algorithms for deriving optimal antenna weights in the time domain, Kohno (2). These algorithms are not going to describe in this paper as they

are less important compare to the DOA algorithms. Moreover, the CMA is not appropriate for the CDMA system because of the required power control, Godara (5).

DOA estimation techniques such as MUSIC, Schmidt (6) and (7), MEM, Nagatsuka et al (8), ESPRIT, Roy et al (9) and DFT, Watanabe et al (10) are the subspace method, from the viewpoint of spectral analysis in the space domain. Some novel methods that provide spatial and temporal estimation such as JADE, Vanderveen (11) and (12), Kohno (2) and hybrid algorithm of DFT and LMS, Watanabe (10), are also been proposed.

Conventional Method for DOA Estimation

Conventional methods for DOA estimation are based on the concepts of beamforming and null-steering and do not exploit the nature of the model of the received signal vector $U(k)$ or the statistical model of the signals and noise. Given the knowledge of the array manifold, an array can be steered electronically. Conventional DOA estimation techniques electronically steer beams in all possible directions and look for peaks in the output power, Schell et al (13). The conventional method discussed here is the delay-and-sum method (classical beamform).

Delay-and-Sum Method The Delay-and-Sum method, also referred to as the Classical beamformer method or Fourier method is one of the simplest techniques for DOA estimation. Fig 7 shows the Classical narrowband beamformer structure, where the output signal $y(k)$ is given by a linearly weighted sum of the sensor element outputs, i.e.

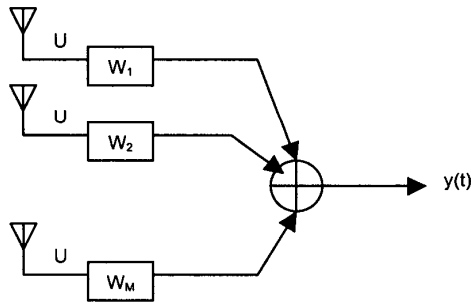


Figure 7 Classical beamforming structure

$$y(k) = W^H U(k)$$

The total output power of the conventional beamformer can be expressed as

Where $\alpha(\theta_0)$ is the steering vector associated with the DOA angle θ_0 , $n(k)$ is the noise vector at the array input, and $\delta_s = E[s(k)^2]$ and $\delta_n = E[n(k)^2]$ are the signal power and noise power, respectively. It is obvious that the output power is maximized when $w = \alpha(\theta_0)$. Therefore of all the possible weight vectors, the receiver antenna has the highest gain in the direction θ_0 , when $w = \alpha(\theta_0)$. This is because the phases of the signal components arriving from θ_0 at the sensors are aligned so they add constructively.

$$P_{\text{avg}}(\theta_0) = \|W^H U(k)\|^2 = E[W^H \alpha(\theta_0) s(k) + n(k)] W$$

$$= |W^H \alpha(\theta_0)|^2 (\delta_s + \delta_n)$$

In the classical beamforming approach to DOA estimation, the beam is scanned over the angular region of interest in discrete steps by forming weights $w =$

$$P_{\text{avg}}(\theta) = W^H R_{UU} W = \alpha^H(\theta) R_{UU} \alpha(\theta)$$

$\alpha(\theta_0)$ for different θ , and the output power is measured. Using the equation above. The output power at the classical beamformer as a function of the AOA is therefore given by:

Therefore, if we have an estimate of the input autocorrelation matrix R_{UU} and know the steering vectors $\alpha(\theta_0)$ for all θ 's of interest (either through calibration or analytical computation), it is possible to estimate the output power as a function of AOA θ . The output power as a function of AOA is often termed the Spatial Spectrum. Clearly, the DOA can be estimated by locating peaks in the spatial spectrum.

The delay-and-sum method has many disadvantages. The width of the beam and the height of the sidelobes limit the effectiveness when the signals arriving from multiple directions and/or sources are present, because the signals over a wide angular region contribute to the measured average power at each look direction. Hence, this technique has poor resolution. Although it is possible to increase the resolution by adding more sensor elements, increasing the number of sensors increases the number of receivers and the amount of storage required for the calibration data, i.e. $\alpha(\theta)$.

Capon's Minimum Variance Method Capon's minimum variance technique, Capon (14) attempts to overcome the poor resolution problem associated with the delay-and-sum method. The technique uses some of the degrees of freedom to form a beam in the desired look direction while simultaneously using the remaining degrees of freedom to form nulls in the direction of interfering signals. This technique minimises the contribution of the undesired interference by minimising the output power while maintaining the gain along the look direction to be constant, usually unity. By computing and plotting Capon's Spectrum over the whole range of θ , the DOA can be estimated by locating the peaks in the spectrum. Though Capon's provides a better resolution when compared to the delay-and-sum method, it method suffers from many disadvantages. It fails if other signals that are correlated with Signal-of-Interest are present because it inadvertently uses that correlation to reduce the processor output power without spatially nulling it, Schell et al (13). Also Capon's method requires the computation of a matrix inverse, which can be expensive for large arrays.

Subspace Methods for DOA Algorithm

Though many of the classical beamforming based methods such as Capon's minimum variance method are often successful and are widely used, these methods have some fundamental limitations in resolution. Most

of these limitations arise because they do not exploit the structure of the input data model. Subspace band methods like MUSIC and ESPRIT exploit the structure of a more accurate data model for the case of sensor arrays of arbitrary form.

Maximum Entropy Method Burg algorithm in MEM uses reflection coefficients at lattice filter in order to estimate the feedback coefficient of an all-pole network. MEM then finds a power spectrum such that its Fourier transform equals the measured correlation subjected to the constraint that its entropy is maximised. The solution to this problem requires an infinite dimensional search, which may be transformed to a finite dimensional search using the duality principle.

Although MEM can accurately estimate arrival angles, there remain the following problems to be solved: (1) Adaptive array antenna based on spatial spectral estimation requires many antenna elements in order to obtain accurate estimates. (2) MEM has larger complexity for implementation compared with DFT. (3) DFT is a linear estimation method, while MEM is non-linear. An improved version of MEM has been proposed in Nagatsuka et al (8), called as MEM Array, which is a method of both improving estimation accuracy and reducing the number of antenna elements. In the method, the arrival angles can be approximately estimated by using temporal sampling instead of spatial sampling.

MUSIC Method The MUSIC algorithm proposed by Schmid (6) and (7) is a high resolution multiple signal classification technique based on exploiting the eigenstructure of the input covariance matrix. MUSIC is a signal parameter estimation algorithm which provides information about the number of incident signals, DOA of each signal, strengths and cross correlation between incident signals, noise power, etc. While the MUSIC algorithm provides very high resolution, it requires very precise and accurate array calibration. The MUSIC algorithm has been implemented and its performance has been experimentally verified. The development of the MUSIC algorithm is based on a geometric view of the signal parameter estimation problem. Following the narrowband data model vector at an M -element array can be expressed as a linear combination of the D incident waveforms and noise. Unlike the conventional methods, the MUSIC Spatial Spectrum does not estimate the signal power associated with each arrival angle. Instead, when the ensemble average of the array input covariance matrix is known exactly, under uncorrelated and identical noise conditions, the peaks of the MUSIC's power spectrum, $P_{\text{MUSIC}}(\theta)$, are guaranteed to correspond to the true DOA. Since these peaks are distinct irrespective of the actual separation between arrival angles, in principle, with perfect array calibration, these estimators can distinguish and resolve arbitrarily closely spaced signals. When the impinging signals $S_i(t)$ are highly correlated, MUSIC fails because the signal correlation matrix R_{ss} becomes singular.

Various modifications to the MUSIC algorithm have been proposed to increase its resolution performance and decrease the computational complexity. One such improvement is the Root-MUSIC algorithm developed and described in Barabell (15), which is based on polynomial rooting and provides high resolution, but it is applicable only to a uniform spaced linear array. Cyclic MUSIC is a signal selective direction finding algorithm, which exploits the spectral coherence of the received signal as well as the spatial coherence. By exploiting spectral correlation along with MUSIC, it is possible to resolve signals spaced more closely than the resolution threshold of the array when only one of them is a Signal-of-Interest (SOI). Cyclic MUSIC also circumvents the requirement that the total number of signals impinging on the array (including both SOI and interference) must be less than the number of sensor elements, Garner (16).

The ESPRIT algorithm is another subspace-based DOA estimation technique developed by Ray et al (9).

Joint Angle and Delay Estimation (JADE)

JADE, Paulraj (1), Schmidt (7) and Vanderveen (12), is a novel subspace-based approach to estimating the Direction-Of-Arrivals (DOAs) and delays of the multipath signals using a collection of space-time channel estimates (Analogous to snap shots in subspace methods for DOA estimation). Which have constant parameters of interest, but different path fade amplitude. A major contribution of the JADE approach is that it can work when the number of paths exceeds the number of antennas, unlike the traditional MUSIC and ESPRIT algorithms. The first step of JADE is estimating the channel (i.e. the impulse response from the user to the antenna array), by using pilot symbol (training bits) or blindly. If the data burst contains a pilot symbol, the channel can be estimated using least squares. We collect data from M consecutive time slots and use it to obtain (noisy) estimates of H . for the sake of simplicity in the simulation, we assume we know the channel H . The final step of JADE is estimating the $2L$ parameters of interest (θ 's and τ 's), and eliminating the L nuisance parameters α 's. JADE-ML model was first investigated and it was modeled and implemented in C++ programming language. But when we arrived the final stage of the search, we were not able to solve the ML optimisation problem to carry on the Newton method to search of the DOA and delay of the signal. And it was confirmed that there is no way to solve this optimisation problem, as it involves a very complex computation and significant amount time is required to carry on. Thus, JADE-MUSIC was then considered.

Since the MUSIC will be employed for solving the JADE equation and perform the search of θ and τ . Hence, it was worthwhile to study the MUSIC in detailed, and implement it by using C++ programming and Matlab. From there we wish to apply MUSIC into JADE and form the JADE-MUSIC.

Fig 8 shows the performance of the MUSIC algorithm for the DOA estimation, where four elements antenna is

employed to collect 16 input samples to estimate the input covariance matrix, followed by eigen (eigenvalues and eigenvectors) decomposition. Finally we calculated the MUSIC spectrum. In the simulation, the single direction of arrival was assumed, i.e. with no multipath present. Fig 8(a) and 8(b) show the result of the MUSIC estimates the signal from direction 60° and 90° respectively. As we can see from the figures, MUSIC can estimate single AOA successfully.

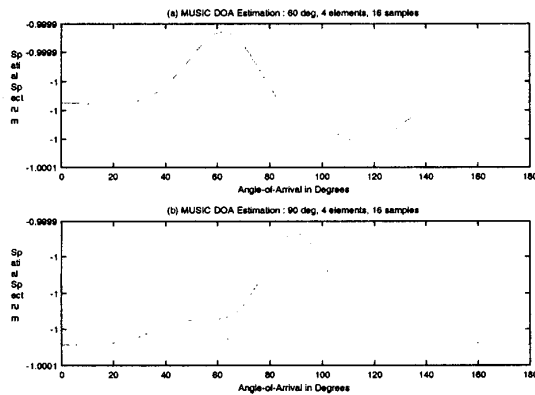


Figure 8 MUSIC DOA estimation for one direction.

In the following simulation a similar assumption is made, i.e. four elements antenna is used with 16 input samples. Due to multipath, two signals arrived at the base station. First, we assume that direct path signal comes from 60° with no delay, followed by a multipath signal from 90° with one chip period delayed. Fig 9 (a) shows that the DOAs estimated by MUSIC is 60° and 100° , instead of 60° and 90° . In other words, MUSIC failed to estimate one of the DOA with a 10° difference, due to the resolution of MUSIC estimation.

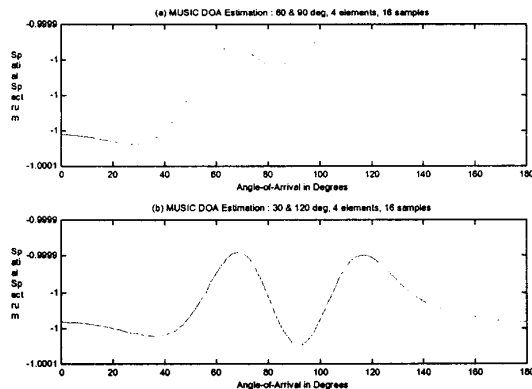


Figure 9 MUSIC DOA estimation for two directions

CONCLUSION

The JADE-Maximum Likelihood was firstly implemented, but due to the inability to solve the Maximum Likelihood optimisation problem, JADE – MUSIC was adopted.

The MUSIC algorithm estimates DOAs in noise subspace, which is defined by eigenvectors of a covariance matrix of spatially sampled signals. The MUSIC has been implemented, and it can estimate one direction of arrival without problem, but it has difficulty in distinguishing two direction-of-arrivals. Since these simulations were based on only 16 and 32 samples, a larger amount of samples can be carried out and the accuracy of the MUSIC estimation should be better. In the simulation, the transmitted signal is assumed to be distorted by fast fading which results in Doppler spread. Hence, if a slow fading channel is assumed, the simulation result would be better.

Since the JADE – MUSIC has not been implemented, it can be considered as one area for further work. A MUSIC based delay estimation algorithm that was proposed in Chu et al (17) can be adopted with the conventional MUSIC algorithm to perform a joint angle and delay estimation technique, i.e. we can estimate both DOA and delay of a multipath signal by using MUSIC based method. This novel hybrid MUSIC method should be more attractive than the JADE method, as MUSIC is considered simple and where the JADE it involves a redundant multi-dimension search, though it works when the number of multipath exceed the number of elements.

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